

Dynamics

Equations

$$\mathbf{s}_f = \mathbf{s}_i + \mathbf{v}_i t + \frac{1}{2} \mathbf{a}_c t^2 \quad \mathbf{v}_f = \mathbf{v}_i + \mathbf{a}_c t$$

$$\mathbf{v}_f^2 = \mathbf{v}_i^2 + 2\mathbf{a}_c(\mathbf{s}_f - \mathbf{s}_i) \quad \int_{\mathbf{s}_i}^{\mathbf{s}_f} \mathbf{a} d\mathbf{s} = \int_{\mathbf{v}_i}^{\mathbf{v}_f} \mathbf{v} d\mathbf{v}$$

$$\int_{\mathbf{s}_i}^{\mathbf{s}_f} d\mathbf{s} = \int_{t_i}^{t_f} \mathbf{v} dt \quad \int_{\mathbf{v}_i}^{\mathbf{v}_f} d\mathbf{v} = \int_{t_i}^{t_f} \mathbf{a} dt$$

$$\mathbf{v}_r = \dot{r} \quad \mathbf{v}_\theta = r\dot{\theta} \quad \mathbf{v}_z = \dot{z}$$

$$|\mathbf{v}| = \sqrt{\mathbf{v}_r^2 + \mathbf{v}_\theta^2 + \mathbf{v}_z^2} \quad \mathbf{a}_z = \ddot{z}$$

$$\mathbf{a}_r = \ddot{r} - r\dot{\theta}^2 \quad \mathbf{a}_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$$

$$|\mathbf{a}| = \sqrt{\mathbf{a}_r^2 + \mathbf{a}_\theta^2 + \mathbf{a}_z^2}$$

$$\mathbf{a}_n = \frac{\mathbf{v}_t^2}{\rho} \quad \mathbf{a}_t = \dot{\mathbf{v}}_t \quad \mathbf{v}_B = \mathbf{v}_A + \mathbf{v}_{B/A}$$

$$\rho = \frac{(1 + (\frac{dy}{dx})^2)^{\frac{3}{2}}}{\frac{d^2y}{dx^2}} \quad \Sigma \mathbf{F} = m\mathbf{a}$$

$$U = \mathbf{F}s \quad \mathbf{F}_s = -k\mathbf{s} \quad V_s, U_s = \int_{s_1}^{s_2} \mathbf{F} ds$$

$$T = \frac{1}{2} m \mathbf{v}^2 + \frac{1}{2} I_G \omega^2 \quad T_1 + \Sigma U_{2-1} = T_2$$

$$T_1 + V_1 = T_2 + V_2 \quad V = mgh + \frac{1}{2} ks^2$$

$$V_g = mgh \quad 1 \text{ hp} \equiv 550 \frac{\text{ft} \cdot \text{lb}}{\text{s}}$$

$$P = \frac{dU}{dt} = \mathbf{F}\mathbf{v} \quad \varepsilon = \frac{\text{power output}}{\text{power input}}$$

$$m\mathbf{v}_i + \int_{t_1}^{t_2} \Sigma \mathbf{F} dt = m\mathbf{v}_f$$

$$m_a \mathbf{v}_{a,i} + m_b \mathbf{v}_{b,i} = m_a \mathbf{v}_{a,f} + m_b \mathbf{v}_{b,f}$$

$$e = \frac{\mathbf{v}_{b,f} - \mathbf{v}_{a,f}}{\mathbf{v}_{a,i} - \mathbf{v}_{b,i}} \quad 0 \leq e \leq 1$$

$$\mathbf{H} = \mathbf{r} \times \mathbf{p} = I\omega \quad \mathbf{r}_i \times \mathbf{p}_i = \mathbf{r}_f \times \mathbf{p}_f$$

$$\mathbf{r}_i \times \mathbf{p}_i + \int \mathbf{r}(t) \times \Sigma \mathbf{F}(t) dt = \mathbf{r}_f \times \mathbf{p}_f$$

$$\mathbf{s} = \theta \mathbf{r} \quad \mathbf{v} = \vec{\omega} \times \mathbf{r} \quad \mathbf{a}_t = \vec{\alpha} \times \mathbf{r}$$

$$\int_{\theta_i}^{\theta_f} \vec{\alpha} d\theta = \int_{\omega_i}^{\omega_f} \vec{\omega} d\omega \quad \int_{\theta_i}^{\theta_f} d\theta = \int_{t_i}^{t_f} \vec{\omega} dt$$

$$\int_{\omega_i}^{\omega_f} d\omega = \int_{t_i}^{t_f} \vec{\alpha} dt \quad I = I_G + md^2$$

$$\mathbf{v}_b = \mathbf{v}_a + \vec{\omega} \times \mathbf{r}_{b/a} \quad \mathbf{v}_a = \mathbf{v}_b + \vec{\omega} \times \mathbf{r}_{a/b}$$

$$\mathbf{a}_b = \mathbf{a}_a + \vec{\alpha} \times \mathbf{r}_{b/a} - \omega^2 \mathbf{r}_{b/a} \quad I_G = mk^2$$

Undamped Free Vibration

Standard equation form:

$$m\ddot{x} + kx = 0$$

if,

$$\omega_n \triangleq \sqrt{\frac{k}{m}}$$

then:

$$\ddot{x} + \omega_n^2 x = 0$$

$$\omega_n = 2\pi f \quad f\tau = 1$$

Simple pendulum period:

$$\tau = \frac{2\pi}{\omega_n} = 2\pi \sqrt{\frac{l}{g}}$$

Confusers

$H \equiv$ angular momentum

$T \equiv$ kinetic energy

$U \equiv$ work

$V \equiv$ potential energy

$\mathbf{v}_{b/a} \equiv$ velocity of b, relative to a

$\mathbf{r}_{b/a} \equiv$ position of b, from a

Coefficient of Restitution Values

if:

$$e = \begin{cases} 1 & \text{the collision is entirely plastic} \\ 0 & \text{the collision is entirely elastic} \end{cases}$$

otherwise, the collision is partially elastic

Unit Vectors

$\hat{u}_n \equiv$ normal to curvature direction

$\hat{u}_t \equiv$ tangential to motion direction

$\hat{u}_\theta \equiv$ transverse direction

$\hat{u}_r \equiv$ radial direction (towards the origin)

Diagram Tips

Free Body Diagram

1. Apply weight forces
2. Determine contact points/surfaces
3. Apply all forces at contact points/surfaces

Kinetic Diagram

1. Assume motion is in the positive directions
2. Apply resistance to motion moment(s)

Equations

Equate the forces/moments present on the Free Body Diagram to those on the Kinetic diagram, as such:

$$\underbrace{\sum \mathbf{F}_x}_{\text{Free Body Diagram}} = \underbrace{\sum \mathbf{F}_x}_{\text{Kinetic Diagram}}$$

$$\underbrace{\sum \mathbf{F}_y}_{\text{Free Body Diagram}} = \underbrace{\sum \mathbf{F}_y}_{\text{Kinetic Diagram}}$$

$$\underbrace{\sum \mathbf{M}_G}_{\text{Free Body Diagram}} = \underbrace{\sum \mathbf{M}_G}_{\text{Kinetic Diagram}}$$

Impulse Diagram

1. Draw the initial conditions momentum diagram
2. Draw the impulse diagram ($\text{FBD} \cdot t$)
3. Draw the final conditions momentum diagram

Equate the three drawn diagrams according to the following equation:

$$\text{Initial Momenta} + \text{Impulse} = \text{Final Momenta}$$